CSV103:ADVANCE LINUX LABORATORY

L:0 T:0 P:2 Credits:1

Course Outcomes: Through this course students should be able to

CO1 :: understand the fundamentals of the Linux command-line interface (CLI)

CO2 :: knowledge the fundamentals of the Linux command-line interface (CLI)

CO3 :: recommend the Linux file system using various CLI commands.

CO4 :: manage user and group permissions using the Linux CLI.

CO5 :: illustrate tasks using Linux shell scripting.

CO6 :: apply Linux commands for system administration tasks.

List of Practicals / Experiments:

Introduction to Linux CLI

- Explore the Linux terminal environment and its components• Familiarize with basic Linux commands for help, information, and directorynavigation• Practice creating, deleting, and renaming files and directories• Learn about file permissions and ownership using the ls comman

File and Directory Management

- Utilize advanced file and directory management techniques• Practice copying, moving, and renaming files and directories• Understand file and directory permissions using the chmod and chowncommands• Learn about symbolic links and their usage• Explore file compression and archiving using the tar and gzip commands

User and Group Management

- Understand the concept of users and groups in Linux• Create, modify, and delete user accounts using the useradd, usermod, anduserdel commands• Manage user groups using the groupadd, groupmod, and groupdelcommands• Understand and set file permissions for users and groups• Practice switching between users using the su and sudo commands

Shell Scripting Fundamentals

- Introduction to shell scripting concepts• Familiarize with basic shell scripting syntax and structure• Learn about variables, data types, and operators in shell scripts• Practice writing simple shell scripts for automating tasks• Explore conditional statements and loops in shell scripting

Advanced Shell Scripting Techniques

- Enhance shell scripting skills with advanced techniques• Learn about functions and parameter passing in shell scripts• Understand file handling and input/output operations in shell scripts• Practice using error handling and debugging techniques• Explore regular expressions for pattern matching and text processin

Linux System Administration Tasks

- Apply Linux commands for system administration tasks• Manage system processes using the ps, top, and kill commands• Schedule tasks using cron jobs and at commands• Monitor system resources and performance using tools like free, top, andhtop• Install and manage software packages using tools like apt, yum, and dnf

Text Books:

1. LINUX SYSTEM ADMINISTRATION by TOM ADELSTEIN, BILL LUBANOVIC, O'REILLY

References:

1. RUNNING LINUX by MATTHIAS KALLE DALHEIMER, O'REILLY

2. UNIX AND LINUX SYSTEM ADMINISTRATION HANDBOOK by EVI NEMETH, GARTH SNYDER, TRENT R. HEIN, BEN WHALEY, DAN MACKIN, O'REILLY